

Charlie Toone

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Professional Summary

Aeronautical Engineering graduate from Loughborough University, inspired to study through building and flying FPV aircraft. Drawn to multidisciplinary engineering problems, with a desire to explore concepts in depth while understanding how they interact within a wider system. Motivated by collaborative environments where technical ideas can be discussed, explored and developed. Keen to build breadth across engineering while developing deeper technical expertise and stronger interpersonal skills, through projects that explore new technologies, advance engineering knowledge and address significant challenges.

Technical Skills

MATLAB / Simulink

Terramechanics Tyre Model | PID control tuning | Data analysis & visualisation

CAD (NX 12 & CATIA 3DX)

NX: Wing structure, piston & connecting rod | CATIA: F1 front wing initial stages

FEA (NX 12)

Wing structure | Bending of hollow beam

CFD (STAR-CCM+)

Mesh sensitivity study and RANS flow around a wing | LES flow around a cylinder

Education

Bachelor of Engineering in Aeronautical Engineering

Loughborough University | 2022 - 2025

2:1, Upper Second-Class Honours

Relevant modules:

- Aircraft Design, Fixed & Rotary Wing Aircraft Performance, Aircraft Stability & Flight Test
- Thermofluids, Incompressible Aerodynamics, Compressible Flow, Intro to CFD
- Turbomachinery & Propulsion, Gas Turbine Design 1

A Levels

Maths (A), Physics (B), Further Maths (B), Chemistry (B) | Selby College | 2020 - 2022

GCSEs

9 GCSEs including Maths (8), English (6), and Double Science (9-8) | Selby High School | 2016 - 2020

Projects

FPV Aircraft Design, Build & Flight

Hardware assembly · Soldering · PID control · Videography · Final Cut Pro X

- Built and flew foam-board RC planes by marking out, cutting, scoring and moulding parts, assembling the structures, and installing electronics. Improved flight characteristics by adjusting the centre of gravity, servo deflection and trim, and installing winglets/strakes, while investigating the effects of different aircraft configurations.
- Installed an FPV camera system to control the planes via live analogue video, prompting an interest in FPV drones and research into components for my first build. Assembled and soldered components, configured the flight controller and flight-tested this build, leading to deeper research into how component selection affected flight characteristics and performance.
- Iterated drone designs and component selection across multiple builds to meet changing objectives, using lighter frames, higher-Kv motors and different radio systems to improve characteristics such as thrust-to-weight ratio and range. Tuned PID and filter settings to reduce oscillations and improve smoothness, while adjusting rate curves to refine input response.
- Flew extensively both solo and alongside others, including racing and chase-style flying, coordinating movements and communicating relative positions. Produced freestyle and cinematic videos, collaborating on 3D-printed GoPro mounts that accommodated ND filters for improved footage.

Terramechanics Tyre Model - Thesis

MATLAB · Simulink · Data visualisation

- Developed a numerical model of tyre–terrain interaction in MATLAB based on Bekker's pressure–sinkage relationship to investigate vertical load and drawbar pull against slip across various sinkages and deformable terrains.
- Refined the model by replacing initial assumptions with more complex phenomena, then integrated it into a vertical and horizontal ¼-car model within Simulink to simulate vehicle velocity and sinkage response to a torque input.
- Presented a 10-minute walkthrough of the project and defended the methodology during questioning from a panel of 3 lecturers.

CFD Analysis

STAR-CCM+ · RANS · LES · Data visualisation

- Conducted a mesh sensitivity study for a finite wing at 10° angle of attack using a RANS high Y+ approach, progressively refining specific regions and monitoring Y+ values and line probe velocity results to establish a balance between accuracy and computational expense.
- Ran simulations at 5°, 10° and 22° using this mesh to calculate force coefficients and compare CFD predictions with experimental results. Produced velocity and wing-tip streamline visualisations across the three angles, analysing changes in stagnation point location and increasing wing-tip vortex energy, as well as increasing adverse pressure gradient leading to flow separation.
- Ran a 0.5 s LES simulation of flow around a cylinder, analysing instantaneous velocity and pressure fields to investigate turbulent flow and dissipation alongside the resulting time-averaged fields. Monitored drag coefficient over time, using its periodic variation to calculate the time-averaged coefficient and identify the frequency of periodic flow.

Wing Structural Analysis & Load Testing

NX 12 · FEA · Clamp and load machine

- Collaborated in a 3-person team to design a balsa wing structure for 1st year students to manufacture and assemble. Took responsibility for the FEA analysis as part of a comparison between hand calculations, FEA predictions, and experimental results from our load test of the manufactured wing.

PC Build for Racing Simulator

Hardware assembly · Windows 11 · MSI BIOS · Troubleshooting

- Researched compatible components to provide 1440p 165fps graphics performance within budget. Assembled components, routed wires, updated the BIOS, configured Windows, and installed drivers.
- Applied knowledge from this build to advise a friend on component selection, assist with assembly, and diagnose a faulty RAM stick by systematically testing working components from my PC.

Certifications

Flight Laboratory Course, Cranfield University, 2024

(National Flying Laboratory Centre, Saab 340B flight test programme)

Took on the role of a flight test observer for 2 flights on board the Saab 340B, interpreting the digital instrument panel as the pilot performed several manoeuvres demonstrating the aircraft's flight characteristics. Analysed the flight test data to evaluate longitudinal and lateral-directional stability, estimate drag, and assess cruise, climb, and take-off performance, comparing results with theoretical values.

Interests & Activities

Polo Player at Loughborough University Polo Club, 2024 – 2025

Learned to ride and play polo through group lessons involving independent drills, paired exercises and practice games led by the coach. Competed in both tournaments, achieving 2nd place in one. Regularly participated in group social events, with my commitment recognised by the club.

Clay Pigeon Committee Member at Loughborough University Clay Shooting, 2023 – 2025

Regularly participated in group social events, joining the committee in 3rd year to help support club socials.

Motorsport & Sim racing, Independent, 2022 – present

Develop racing skills through force-feedback simulator practice. Attend F1 and other motorsport events.

FPV Aircraft Builder & Pilot, Independent, 2017 – present

Design, build and pilot RC FPV aircraft, capturing on-board footage to create freestyle and cinematic videos.

Full UK driving licence | Languages: English - native, Spanish - basic | Proficient in Microsoft Office